

- 12 (a) (i) picking up of signal in one cable from a second (nearby) cable M1
A1 [2]
- (ii) random (unwanted) signal / power B1
that masks / added to / interferes with / distorts transmitted signal B1 [2]
(allow this mark in (i) or (ii))
- (b) if P is power at receiver, C1
 $30 = 10\lg(P / (6.5 \times 10^{-6}))$ C1
 $P = 6.5 \times 10^{-3} \text{ W}$ C1
loss along cable = $10\lg(\{26 \times 10^{-3}\} / \{6.5 \times 10^{-3}\})$ C1
= 6.0 dB C1
length = $6.0 / 0.2 = 30 \text{ km}$ A1 [5]